

EXPLORATORY DATA ANALYSIS ON TITANIC DATASET

Introduction

The Titanic Dataset is popular for data analysis and machine learning. It contains information about the passengers on-board the Titanic, including features like age, gender, fare, cabin, and survival status. We will perform exploratory data analysis on the Titanic dataset using Python in this project.

Dataset

The Titanic dataset consists of the following columns:

- Survived: Survival status(0=No, 1=Yes)
- Pclass: Passenger class(1=1st class, 2=2nd class, 3=3rd class)
- Sex: Passenger's gender
- Age: Passenger's age
- SibSp: Number of siblings/spouses aboard
- Parch: Number of parents/children aboard
- Fare: Fare paid for the ticket
- Embarked: Port of embarkation

Project Steps

1. Importing Libraries:

Starting by importing the required libraries for data analysis and visualization:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

2. Loading the Dataset:

Now load the dataset i.e. Titanic-Dataset directly into the DataFrame:

```
df = pd.read_csv('Titanic-Dataset.csv')
```

3. Exploring the Data:

To gain initial insights into the dataset, we can perform some basic exploratory operations:

```
print("Shape of Dataset:", df.shape)
print("\n Data Info:")
df.info()
```

```
print("\n Statistical Summary:")
print(df.describe(include='all'))
```

```
print("\n Missing Values:")
print(df.isnull().sum())
```

4. Data Visualization:

Visualization helps us understand the data and identify patterns. We can create various types of plots using libraries like Matplotlib and Seaborn. Here are some examples:

```
sns.pairplot(df[['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare']], hue='Survived', palette='Set1')
plt.suptitle("Pairplot-Survival & Features", y=1.02)
plt.show()
```

```
df.hist(bins=30, figsize=(15,10), color='skyblue')
plt.suptitle("Histograms of Numeric Features")
plt.show()
```

```
sns.boxplot(x='Pclass', y='Fare', data=df, palette='Pastel1')
plt.title("Boxplot:Fare by class")
plt.show()
```

```
sns.boxplot(x='Survived', y='Age', data=df, palette='Set2')
plt.title("Boxplot:Age by Survival")
plt.show()
```

```
sns.countplot(x='Survived', data=df, palette='Set1')
plt.title("Count Plot:Survival")
plt.show()
```

```
sns.countplot(x='Pclass', data=df, palette='Set1')
```

5. Exploring Relationships:

EDA involves exploring relationships between variables to uncover insights.

```
sns.heatmap(df.corr(numeric_only=True),annot=True,cmap="coolwarm",fmt=".2f")  
plt.title("Correlation Heatmap")  
plt.show()
```

6. Conclusion:

Based on the exploratory data analysis, we can summarize the key findings, insights, and potential areas for further investigation. This includes patterns, trends, outliers, or relationships observed during the analysis.